Performance Testing

Date	15 March 2026
Team ID	unm293200909
Project Name	FitBuddy - AI Fitness Plan Generator using Gemini Models
Name	Kavyasri.A
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how FitBuddy behaves under expected and peak load conditions. The sections below document the testing approach, tool used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust (Python load testing tool)
Type of Testing	Load Testing
Target Module / API	GET /, POST /generate-workout, POST /submit-feedback, GET /view-all-users
Test Environment	Local (Uvicorn server at http://127.0.0.1:8000)
Test Date	*[Enter the date you run the tests]*

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Home page load - GET /	**20**	**60**	Page loads in under 2 seconds with no errors
2	Workout plan generation - POST /generate-workout (Gemini 1.5 Pro and Flash)	**5**	**60**	Plan and tip returned in under 5 seconds; data saved in SQLite
3	Feedback-based plan update - POST /submit-feedback	**5**	**60**	Updated plan returned in under 5 seconds
4	Admin dashboard - GET /view-all-users	**10**	**60**	Users and plans listed in under 2 seconds

Step 3: Performance Test Results

Actual values, status and remarks are to be filled in after running the tests.

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds			
2	Response Time (Max)	< 5 seconds			
3	Throughput (Req/sec)	At least 1			
4	Error Rate	< 1%			
5	CPU Utilization	< 80%			
6	Memory Utilization	< 80%			

Step 4: Observations & Analysis

Key Findings:
[Describe what the test results revealed about FitBuddy's performance]

Bottlenecks Identified:
[List bottlenecks observed. Area to check: Gemini API response time on /generate-workout and /submit-feedback]

Optimization Steps Taken:
[Describe changes made, e.g. shorter prompts, a faster model for simple tasks]

Step 5: Screenshots / Evidence

Attach screenshots of the Locust results (statistics table and charts) below:

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]

Work Assignment:

Work Assigned	Assigned To
Install Locust and write the test script (locustfile.py)	Kavyasri.A
Run the four test scenarios and record the results	Kavyasri.A
Analyse results, identify bottlenecks and optimise	Kavyasri.A
Capture screenshots and complete this report	Kavyasri.A